

Protein aggregation in bacteria: the thin boundary between functionality and toxicity.

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Misfolding and aggregation of proteins have a negative impact on all living organisms. In recent years, aggregation has been studied in detail due to its involvement in neurodegenerative diseases, including Alzheimer's, Parkinson's and Huntington's diseases, and type II diabetes--all associated with accumulation of amyloid fibrils. This research highlighted the central importance of protein homeostasis, or proteostasis for short, defined as the cellular state in which the proteome is both stable and functional. It implicates an equilibrium between synthesis, folding, trafficking, aggregation, disaggregation and degradation. In accordance with the eukaryotic systems, it has been documented that protein aggregation also reduces fitness of bacterial cells, but although our understanding of the cellular protein quality control systems is perhaps most detailed in bacteria, the use of bacterial proteostasis as a drug target remains little explored. Here we describe protein aggregation as a normal physiological process and its role in bacterial virulence and we shed light on how bacteria defend themselves against the toxic threat of aggregates. We review the impact of aggregates on bacterial viability and look at the ways that bacteria use to maintain a balance between aggregation and functionality. The proteostasis in bacteria can be interrupted via overexpression of proteins, certain antibiotics such as aminoglycosides, as well as antimicrobial peptides--all leading to loss of cell viability. Therefore intracellular protein aggregation and disruption of proteostatic balance in bacteria open up another strategy that should be explored towards the discovery of new antimicrobials.

[Neurodegenerative amyloidoses: the yeast model].

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More than 20 human diseases are related to protein misfolding which causes formation of amyloids, fibrillar aggregates of normally soluble proteins. Such diseases are called amyloid diseases or amyloidoses. Of them only prion diseases are transmissible. Amyloids of the prion type are described in lower eukaryotes. However, in contrast to mammalian prions, which cause incurable neurodegenerative diseases, prions of lower eukaryotes are related to some non-chromosomally inherited phenotypic traits. Here we summarize the results of studies of prions of the yeast *Saccharomyces cerevisiae* and of the use of yeast model for investigation of some human amyloidoses, such as prion diseases, Alzheimer's, Parkinson's, and Huntington's diseases.

Neurodegenerative disorders: dysregulation of a carefully maintained balance?

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The aggregation of misfolded proteins has long been regarded as a pathological event in neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease and Huntington's disease. However, the exact molecular mechanisms that govern protein metabolism that may lead to toxicity remain largely unclear. Originally targeted as the causative agent, it has since become evident that aggregation formation may not be necessary for disease progression and studies show that they may even serve functional and protective roles. Although the focus has since shifted to the toxicity of intermediate protein species preceding aggregation formation, many questions remain: Is the blame for

the neural destruction to be put on one event alone, or rather on a state of cellular disequilibrium resulting from multiple events? If the cause is multifactorial, then what triggers the toxic cascade and how can this be targeted therapeutically? In order to understand the origin of toxicity, the exact underlying mechanism and impact of each contributing process must be assessed. Therefore, the structural properties, mechanism of formation, cytotoxic and/or protective effects, as well as the clinical impact of protein intermediates and aggregates will be reviewed here with the goal to establish a neurodegenerative disease model aimed at improving current therapeutics, which may ultimately contribute towards improved treatment modalities.

Protein misfolding and aggregation in neurodegenerative diseases: a review of pathogeneses, novel detection strategies, and potential therapeutics.

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Protein folding is a complex, multisystem process characterized by heavy molecular and cellular footprints. Chaperone machinery enables proper protein folding and stable conformation. Other pathways concomitant with the protein folding process include transcription, translation, post-translational modifications, degradation through the ubiquitin-proteasome system, and autophagy. As such, the folding process can go awry in several different ways. The pathogenic basis behind most neurodegenerative diseases is that the disruption of protein homeostasis (i.e. proteostasis) at any level will eventually lead to protein misfolding. Misfolded proteins often aggregate and accumulate to trigger neurotoxicity through cellular stress pathways and consequently cause neurodegenerative diseases. The manifestation of a disease is usually dependent on the specific brain region that the neurotoxicity affects. Neurodegenerative diseases are age-associated, and their incidence is expected to rise as humans continue to live longer and pursue a greater life expectancy. We presently review the sequelae of protein misfolding and aggregation, as well as the role of these phenomena in several neurodegenerative diseases including Alzheimer's disease, Huntington's disease, amyotrophic lateral sclerosis, Parkinson's disease, transmissible spongiform encephalopathies, and spinocerebellar ataxia. Strategies for treatment and therapy are also conferred with respect to impairing, inhibiting, or reversing protein misfolding.

Protein misfolding and aggregation in ageing and disease: molecular processes and therapeutic perspectives.

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Although intensively researched, the fundamental mechanism of protein misfolding that leads to protein aggregation and associated diseases remains somewhat enigmatic. The failure of a protein to correctly fold de novo or to remain correctly folded can have profound consequences on a living system especially when the cellular quality control processes fail to eliminate the rogue proteins. Over 20 different human diseases have now been designated as 'conformational diseases' and include neurodegenerative diseases such as Alzheimer's disease (AD), Huntington's disease (HD) and Creutzfeldt Jakob disease (CJD) that are becoming increasingly prevalent in an ageing human population. Such diseases are usually characterised by the deposition of specific misfolded proteins as amyloid fibrils and hence are often referred to as the amyloidoses.

Basic mechanisms of neurodegeneration: a critical update.

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Neurodegenerative diseases are characterized by progressive dysfunction of specific populations of neurons, determining clinical presentation. Neuronal loss is associated with extra and intracellular accumulation of misfolded proteins, the hallmarks of many neurodegenerative proteinopathies. Major basic processes include abnormal protein dynamics due to deficiency of the ubiquitin-proteasome-autophagy system, oxidative stress and free radical formation, mitochondrial dysfunction, impaired bioenergetics, dysfunction of neurotrophins, 'neuroinflammatory' processes and (secondary) disruptions of neuronal Golgi apparatus and axonal transport. These interrelated mechanisms lead to programmed cell death is a long run over many years. Neurodegenerative disorders are classified according to known genetic mechanisms or to major components of protein deposits, but recent studies showed both overlap and intraindividual diversities between different phenotypes. Synergistic mechanisms between pathological proteins suggest common pathogenic mechanisms. Animal models and other studies have provided insight into the basic neurodegeneration and cell death programs, offering new ways for future prevention/treatment strategies.

Neurodegenerative Diseases: From Dysproteostasis, Altered Calcium Signaling to Selective Neuronal Vulnerability to AAV-Mediated Gene Therapy.

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Despite intense research into the multifaceted etiology of neurodegenerative diseases (ND), they remain incurable. Here we provide a brief overview of several major ND and explore novel therapeutic approaches. Although the cause (s) of ND are not fully understood, the accumulation of misfolded/aggregated proteins in the brain is a common pathological feature. This aggregation may initiate disruption of Ca^{++} signaling, which is an early pathological event leading to altered dendritic structure, neuronal dysfunction, and cell death. Presently, ND gene therapies remain unidimensional, elusive, and limited to modifying one pathological feature while ignoring others. Considering the complexity of signaling cascades in ND, we discuss emerging therapeutic concepts and suggest that deciphering the molecular mechanisms involved in dendritic pathology may broaden the phenotypic spectrum of ND treatment. An innovative multiplexed gene transfer strategy that employs silencing and/or over-expressing multiple effectors could preserve vulnerable neurons before they are lost. Such therapeutic approaches may extend brain health span and ameliorate burdensome chronic disease states.

The role of microRNA-485 in neurodegenerative diseases.

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Neurodegenerative diseases (NDDs) are age-related disorders characterized by progressive neurodegeneration and neuronal cell loss in the central nervous system. Neuropathological conditions such as the accumulation of misfolded proteins can cause neuroinflammation, apoptosis, and synaptic dysfunction in the brain, leading to the development of NDDs including Alzheimer's disease (AD) and Parkinson's disease (PD). MicroRNAs (miRNAs) are small noncoding RNA molecules that regulate gene

expression post-transcriptionally via RNA interference. Recently, some studies have reported that some miRNAs play an important role in the development of NDDs by regulating target gene expression. MiRNA-485 (miR-485) is a highly conserved brain-enriched miRNA. Accumulating clinical reports suggest that dysregulated miR-485 may be involved in the pathogenesis of AD and PD. Emerging studies have also shown that miR-485 plays a novel role in the regulation of neuroinflammation, apoptosis, and synaptic function in the pathogenesis of NDDs. In this review, we introduce the biological characteristics of miR-485, provide clinical evidence of the dysregulated miR-485 in NDDs, novel roles of miR-485 in neuropathological events, and discuss the potential of targeting miR-485 as a diagnostic and therapeutic marker for NDDs.

Nanoparticles in Combating Neuronal Dysregulated Signaling Pathways: Recent Approaches to the Nanoformulations of Phytochemicals and Synthetic Drugs Against Neurodegenerative Diseases.

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As the worldwide average life expectancy has grown, the prevalence of age-related neurodegenerative diseases (NDDs) has risen dramatically. A progressive loss of neuronal function characterizes NDDs, usually followed by neuronal death. Inflammation, apoptosis, oxidative stress, and protein misfolding are critical dysregulated signaling pathways that mainly orchestrate neuronal damage from a mechanistic point. Furthermore, in afflicted families with genetic anomalies, mutations and multiplications of α -synuclein and amyloid-related genes produce some kinds of NDDs. Overproduction of such proteins, and their excessive aggregation, have been proven in various models of neuronal malfunction and death. In this line, providing multi-target therapies carried by novel delivery systems would pave the road to control NDDs through simultaneous modulation of such dysregulated pathways. Phytochemicals are multi-target therapeutic agents, which employ several mechanisms towards neuroprotection. Besides, the blood-brain barrier (BBB) is a critical issue in managing NDDs since it inhibits the accessibility of drugs to the brain in sufficient concentration. Besides, discovering novel delivery systems is vital to improving the efficacy, bioavailability, and pharmacokinetic of therapeutic agents. Such novel formulations are also employed to improve the drug's biodistribution, allow for the co-delivery of several medicines, and offer targeted intracellular delivery against NDDs. The present review proposes nanoformulations of phytochemicals and synthetic agents to combat NDDs by modulating neuroinflammation, neuroapoptosis, neuronal oxidative stress pathways and protein misfolding.

Roles of Oxidative Stress in Synaptic Dysfunction and Neuronal Cell Death in Alzheimer's Disease.

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Alzheimer's disease (AD) is a brain disorder that progressively undermines memory and thinking skills by affecting the hippocampus and entorhinal cortex. The main histopathological hallmarks of AD are the presence of abnormal protein aggregates (A β and tau), synaptic dysfunction, aberrant proteostasis, cytoskeletal abnormalities, altered energy homeostasis, DNA and RNA defects, inflammation, and neuronal cell death. However, oxidative stress or oxidative damage is also evident and commonly overlooked or considered a consequence of the advancement of dementia symptoms. The control or

onset of oxidative stress is linked to the activity of the amyloid- β peptide, which may serve as both antioxidant and pro-oxidant molecules. Furthermore, oxidative stress is correlated with oxidative damage to proteins, nucleic acids, and lipids in vulnerable cell populations, which ultimately lead to neuronal death through different molecular mechanisms. By recognizing oxidative stress as an integral feature of AD, alternative therapeutic or preventive interventions are developed and tested as potential or complementary therapies for this devastating neurodegenerative disease.
